

Monetary Transmission under Competing Identification Schemes

Euro area vs Türkiye

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<https://github.com/hakangulmez/macro-var-lab>

Abstract

We compare monetary-response methods on a common euro-area sample with fixed lag orders three and seven. Recursive and sign-restricted VARs, recursive-shock local projections and proxy VARs use identical variables, transformations and shock normalization within each panel. At twelve months, proxy euro appreciation is 4.551 and 3.305 log percent respectively, with conditional bands excluding zero. Proxy price estimates are 0.172 and -0.078, both imprecise. Türkiye's inherited recursive price estimate is -0.288, also imprecise. Exogeneity is assumed, proxy uncertainty conditions on fitted slopes, and slightly unstable dynamics limit causal interpretation.

1 Introduction and contribution

Identifying the effect of monetary tightening requires distinguishing policy innovations from endogenous responses to prices, activity and financial conditions. A change in a market yield is not automatically a policy shock. This distinction matters particularly when the policy indicator embeds expectations or when an effective funding rate differs from the announced rate.

The literature supplies complementary approaches. Sims et al. (1990) study inference in linear systems with unit roots; their analysis motivates retaining a levels specification while treating integration diagnostics as qualifications, not mechanical differencing instructions. Jordà (2005) estimates responses horizon by horizon. Uhlig (2005) demonstrates identification through sign restrictions, which need not deliver a uniquely identified model. Kilian (1998) studies small-sample impulse-response intervals. Altavilla et al. (2019) provide high-frequency euro-area monetary-event measurements useful as external instruments.

This exercise contributes a transparent empirical comparison rather than a new identification theorem. We use the same empirical estimates throughout the tables and figures. Responses are normalized to one percentage point of the policy indicator and followed for 36 months. The comparison also documents transformations, lag-screening rules, incomplete diagnostics and rejected alternatives. Our methods comparison on a common sample and lag specification holds the variables, transformations, deterministic terms and normalization fixed within each lag panel. It changes both shock identification and, for recursive-shock LP, the response estimator.

2 Data

2.1 Official series and baseline samples

The euro-area baseline covers 2004-09–2026-07; Türkiye covers 2011-01–2026-07. The former uses an ECB fitted AAA two-year spot yield, the latter CBRT’s average funding cost (AOFM). The yield is a market-based indicator, not an overnight administered rate. Industrial production is seasonally/calendar adjusted; price indices are not, so monthly deterministic indicators enter the model. Euro-area geography follows the source’s EA21 series rather than a reconstructed historical membership panel. The table records available source coverage. The actual model windows are

Table 1: Series definitions: baseline inputs

Variable	Source / exact ID and selectors	Transformation	Observed coverage
ea_ip	eurostat: sts_inpr_m; PRD, B-D, SCA, I21, EA21	100 log(level)	1999-01 to 2026-07
ea_hicp	eurostat: prc_hicp_minr; I25, TOTAL, EA21	100 log(level)	1999-12 to 2026-09
ea_2y	ecb: YC.B.U2.EUR.4F.G_N_A.SV_C_YM.SR_2Y	pp levels	2004-09 to 2026-09
eurusd	fred: DEXUSEU	100 log(level)	1999-01 to 2026-09
brent	fred: DCOILBRENT	100 log(level)	1999-01 to 2026-09
vix	fred: VIXCLS	log(level)	1999-01 to 2026-09
tr_ip	evds: TP.TSANAYMT2021.BCD	100 log(level)	2006-01 to 2026-07
tr_cpi	evds: TP.TUKFIY2025.GENEL	100 log(level)	2006-01 to 2026-09
tr_aofm	evds: TP.APIFON4	pp levels	2011-01 to 2026-09
usdtry	evds: TP.DK.USD.A.YTL	100 log(level)	2006-01 to 2026-09
fedfunds	fred: FEDFUNDS	pp levels	1999-01 to 2026-09

Source: author’s calculations from the documented inputs. Coverage is source availability, not a common estimation sample. Daily series use observed monthly means before transformation. Missing months remain missing.

fixed above and are not extended to the last available value of each component. We use the acquisition vintages recorded in the source metadata. The new comparison uses 2005-04–2025-10, 247 usable observations at both fixed lags. Seven earlier months supply lags only; the shorter-lag fit is restricted to the same start. This mechanically reflects the frozen yield availability, not specification search. We retain the source observations privately and document series identifiers, terms and attribution in the data documentation.¹

2.2 Alternative indicators and construction checks

We construct the announced Türkiye comparator from the CBRT overnight borrowing rate before the change to the one-week repo indicator, then the one-week repo lending rate. We average daily observations by month and do not backfill AOFM with the announced-rate comparator. The longer sample is an explicitly labelled robustness exercise.

We use EONIA before the transition and €STR thereafter, with a transition-level dummy. A separately constructed adjusted-EONIA series is not substituted into these model specifications. The Bundesbank German zero-coupon two-year yield is a German long-sample alternative, not a substitute labelled as an ECB policy rate. EA21 price coverage leaves 11 unavailable early months; long-sample price specifications therefore begin at the first available month.

Positive indices, industrial production, exchange rates and Brent enter as 100 log levels; VIX enters as an unscaled log and rates as percentage-point levels. We average daily observations

¹The repository file `DATA.md` records source terms and acquisition details; FRED acquisition uses its official observations API.

Table 2: Series definitions: declared alternatives

Variable	Source / exact ID and selectors	Transformation	Observed coverage
ea_core	eurostat: prc_hicp_minr; I25, TOT_X_NRG_FOOD, EA21	100 log(level)	1999-12 to 2026-09
ea_eonia	ecb: FM.M.U2.EUR.4F.MM.EONIA.HSTA	pp levels	1999-01 to 2021-12
ea_estr	ecb: EST.B.EU000A2X2A25.WT	pp levels	2019-10 to 2026-09
de_2y	bundesbank: BBSIS.D.I.ZST.ZI.EUR.S1311.B.A604.R02XX.R.A.A._Z._Z.A	pp levels	1999-01 to 2026-09
tr_core_b	evds: TP.FE25.OKTG03	100 log(level)	2006-01 to 2026-09
tr_core_c	evds: TP.FE25.OKTG04	100 log(level)	2006-01 to 2026-09
tr_reer	evds: TP.RK.T1.Y	100 log(level)	2006-01 to 2026-09
tr_cpi_old	evds: TP.GENENDEKS.T1	100 log(level)	2006-01 to 2026-09
tr_core_b_old	evds: TP.FE.OKTG03	100 log(level)	2006-01 to 2025-12
tr_core_c_old	evds: TP.FE.OKTG04	100 log(level)	2006-01 to 2025-12
tr_announced	CBRT public effective-date O/N borrowing and one-week repo lending tables	pp; daily step mean	2006-01 to 2026-09

Source: author’s calculations from the documented inputs. Coverage is source availability, not a common estimation sample. Daily series use observed monthly means before transformation. Missing months remain missing.

before logging. FX is USD per EUR for the euro area and TRY per USD for Türkiye: increases mean appreciation and depreciation respectively. No missing value is interpolated. Current-base Turkish core indices are used without splicing; the incompatible old/current rebasing diagnostics are retained in the appendix.

Table 3: Price-rebasing bridge diagnostics

Index	Overlap	Growth gap pp	Ratio range %	Tolerance %
headline	2005-01 / 2026-09	0.006544	0.007855	0.200
core_b	2005-01 / 2025-12	0.199831	0.200031	0.200
core_c	2005-01 / 2025-12	0.146075	0.207526	0.200

Source: author’s calculations from the documented inputs. Current series already cover the analysis window. No core-index splice is used.

3 Econometric model

3.1 Reduced-form system, deterministics and lag choice

For the euro area, the ordered vector is $(\log B, \log V, \log IP, \log P, r, \log FX)'$, with the scaling described above. Türkiye inserts the Federal funds rate after VIX and before domestic activity. Let d_t contain an intercept and monthly indicators, plus an explicitly declared transition or easing dummy where applicable. The estimated system is

$$y_t = C d_t + \sum_{\ell=1}^p A_{\ell} y_{t-\ell} + u_t, \quad \widehat{\Sigma} = T_u^{-1} \sum_t \widehat{u}_t \widehat{u}_t'. \quad (1)$$

We estimate each equation by OLS on its permitted regressors. In Türkiye, our external block (Brent, VIX, Federal funds) excludes all lags of domestic variables from its own equations; domestic equations retain all lags. These lag exclusions do not turn the system into a model with contemporaneously exogenous foreign innovations. The covariance uses the common maximum-likelihood divisor T_u , preserving positive definiteness across different equation masks.

Candidate information criteria use the same feasible-lag comparison window:

$$IC_{AIC}(p) = \log |\widehat{\Sigma}_p| + 2q_p/T_c, \quad (2)$$

$$IC_{BIC}(p) = \log |\widehat{\Sigma}_p| + q_p \log T_c/T_c, \quad (3)$$

$$IC_{HQ}(p) = \log |\widehat{\Sigma}_p| + 2q_p \log \log T_c/T_c. \quad (4)$$

Here q_p counts allowed equation coefficients, including deterministics. Starting at the BIC choice, we increase the lag until every marginal equation BG test at the first residual lag has $p \geq .05$, up to the fixed cap 12. If none passes, the BIC model is retained with an explicit failure flag. Selected models are refit using all observations available after their own lag. Inherited baseline choices are 7 and 9. The new common-sample comparison fixes $p = 3$ and $p = 7$ without running that lag-selection rule; both use identical residual dates.

We retain a levels baseline for the euro area. For Türkiye, we first estimate levels; we switch the price transformation if the companion spectral radius is at least one or the first-difference integration agreement rule is I(2)-compatible. The saved levels radius is 1.01298, and our baseline therefore uses monthly price inflation. We do not subsequently change the other transformations to force stationarity. Stability concerns remain part of interpretation.²

3.2 Structural identification and propagation

We set $\Phi_0 = I$ and recursively define $\Phi_h = \sum_{\ell=1}^{\min(p,h)} A_\ell \Phi_{h-\ell}$. If $\widehat{\Sigma} = LL'$ with lower-triangular L , recursive innovations satisfy $\epsilon_t = L^{-1}u_t$. For policy position j ,

$$b_j = Le_j/L_{jj}, \quad IRF_h = \Phi_h b_j. \quad (5)$$

The shock raises the rate by one percentage point on impact. Ordered predecessors cannot respond contemporaneously under this recursive restriction. Türkiye's FX-first alternative swaps the final policy and FX positions; it leaves the prior equations unchanged. Identification changes impact exclusions, not merely the plot order.

For sign identification, we hold the estimated reduced form fixed. We draw a marginal orthogonal column using a normalized Gaussian vector q , with Türkiye's external coordinates set to zero. We orient the vector so that the policy impact is positive. We admit rotations only if

$$(\Phi_h Lq)_j \geq 0, \quad (\Phi_h Lq)_P \leq 0 \quad (h = 0, 1, 2, 3), \quad (Lq)_j \geq 0.1\sqrt{\widehat{\Sigma}_{jj}}. \quad (6)$$

For inflation-native prices, the price condition instead applies to the cumulative price path. We normalize each accepted rotation by its rate impact. Medians and minima/maxima across accepted rotations summarize a finite admissible set. They are not sampling confidence intervals; coordinate medians need not constitute one admissible structural model.

We construct the proxy from monthly sums of event-window yield changes in EA-MPD, over 2004-09–2025-10.³ Known non-event months inside coverage are zero; later months are missing. With centered instrument z_t and residuals,

$$\widehat{b}^{IV} = \frac{\widehat{\text{Cov}}(z_t, u_t)}{\widehat{\text{Cov}}(z_t, u_{jt})}, \quad IRF_h^{IV} = \Phi_h \widehat{b}^{IV}. \quad (7)$$

²We implement OLS estimation, propagation and historical accounting in `var.py`; lag screening, bias correction, cumulative local projections and proxy inference in the monetary-inference module; HAC inference and sign rotations in `identification.py`; and transformation criteria and recent-regime projections in the model-specification module. The source files specify the procedures described in the model section.

³We use the `OIS_2Y` field in the combined `MonetaryEventWindow`.

The rate residual first stage includes centering, conventional homoskedastic slope variance and $F = \hat{\pi}^2 / \widehat{\text{Var}}(\hat{\pi})$. We prefer the proxy estimate when $F > 10$ and the policy covariance is non-negligible. The common-sample first-stage statistics are $F = 19.598$ at $p = 3$ and $F = 18.826$ at $p = 7$, each on 247 residual observations. Both meet the fixed relevance condition, but neither proves exclusion or separates information shocks. No replacement instrument or lag is searched.

3.3 Local projections and inference

The recursive rate-unit innovation is $s_t = L_{jj}\epsilon_{jt}$. At each horizon, we partial the shock and outcome off W_t , consisting of the VAR design and contemporaneous ordered predecessors. Let tildes denote those residuals. The projection coefficient and its score are

$$\hat{\beta}_h = \frac{\sum_t \tilde{s}_t \tilde{y}_{t+h}}{D_h}, \quad D_h = \sum_t \tilde{s}_t^2, \quad a_{t,h} = \tilde{s}_t (\tilde{y}_{t+h} - \hat{\beta}_h \tilde{s}_t). \quad (8)$$

For differenced prices, the dependent outcome is the cumulative price index, with its pre-shock level added as a nuisance regressor. The price-level bands therefore come from a direct regression, not a sum of standard errors. Horizon-specific Bartlett HAC variance uses $b_h = \min(h+1, n_h-1)$:

$$\widehat{\text{Var}}(\hat{\beta}_h) = \frac{n_h}{n_h - r_h - 1} \frac{\sum_t a_{t,h}^2 + 2 \sum_{\ell=1}^{b_h} (1 - \ell/(b_h+1)) \sum_t a_{t,h} a_{t-\ell,h}}{D_h^2}. \quad (9)$$

We require adequate residual degrees of freedom and leave unsupported horizons unreported. External-block responses are set to zero by the LP reporting restriction. The recent Türkiye regime uses only domestic outcomes and one lag, retaining the full-sample retrospectively identified shock and normalizing its recent policy impact. No full recent-regime VAR is estimated.

We construct recursive pointwise percentile bands using the Kilian-style bootstrap-after-bootstrap: 200 first-stage draws estimate coefficient bias, and 499 outer draws reuse that bias. The coefficient-bias step is $\hat{B} = \hat{A}^* - \hat{A}$, giving candidate generator $\tilde{A} = \hat{A} - w\hat{B}$ with our stationarity-preserving shrink weight w . Residual vectors are centered and sampled independently in this branch. A stable starting model receives bias correction with shrinkage if correction destabilizes it; an initially nonstationary model remains unadjusted. Hence the label does not imply successful stationarity correction for every draw.

Ordinary circular residual-vector blocks of length 6 provide the separate robustness, FEVD and historical-decomposition bands. Those draws refit coefficients and identification while conditioning on chosen lags and transforms. For common-sample proxy bands, paired residual/instrument block draws yield 407 accepted at $p = 3$ and 379 at $p = 7$, out of 499 at each lag; weak draws are omitted and counted, and propagation slopes remain fixed. LP bands are normal HAC intervals. All are pointwise, not simultaneous or weak-IV-robust sets. We retain both central coverage levels in the supplementary result tables.

4 Results

4.1 Euro area: methods on the common sample

At twelve months proxy euro appreciation is 4.551 [1.889,7.630] at $p = 3$ and 3.305 [0.648,5.505] at $p = 7$. The FX channel remains most clearly measured under this conditional design. Proxy price responses are 0.172 [-2.316,1.895] and -0.078 [-3.132,2.392]; neither pins down the sign. Activity responses are -0.403 and -0.587, likewise imprecise. We retain both lag panels and all methods rather than selecting a stronger narrative. Positive recursive/LP price responses and sign-conditioned early disinflation remain plainly visible.

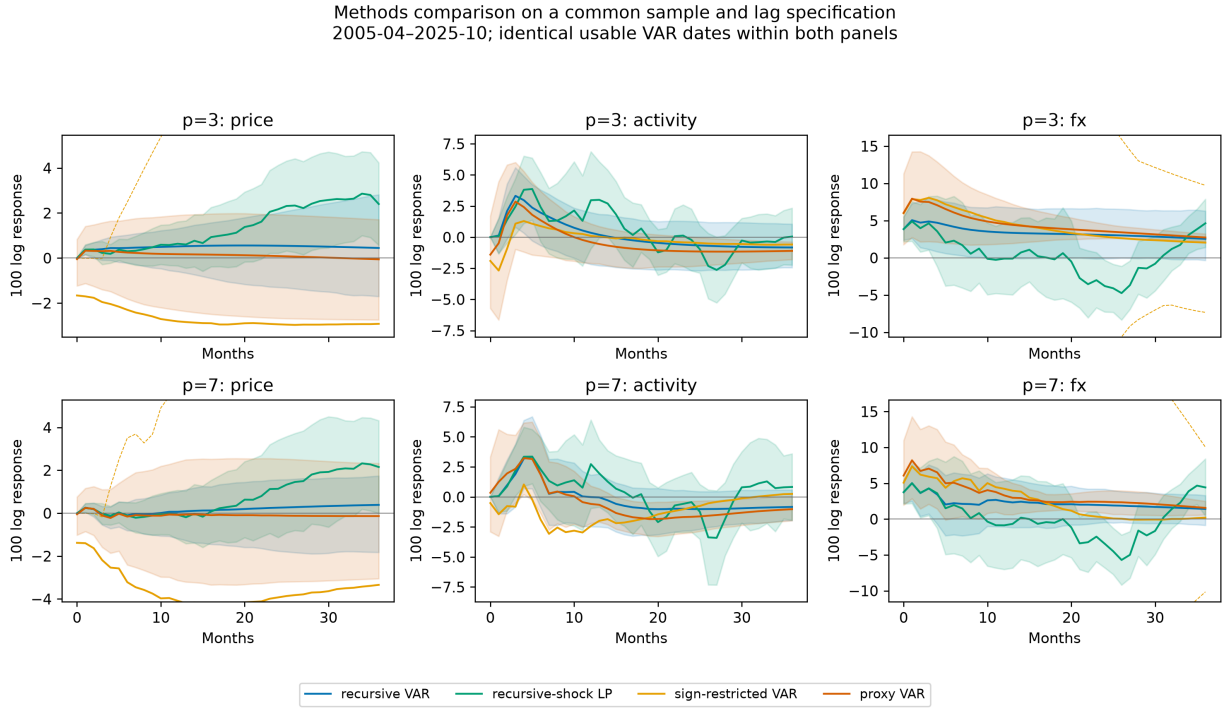
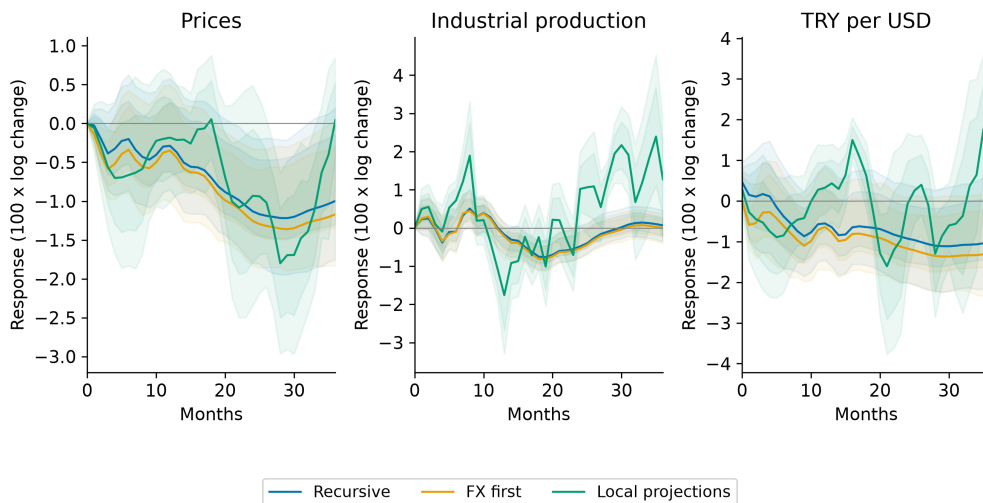


Figure 1: Methods comparison on a common sample and lag specification, with separate fixed-lag panels. Shading is pointwise sampling uncertainty; thin dashed sign extrema are admissible ranges. Proxy bands condition on fixed propagation slopes and omit weak bootstrap draws. LP is explicitly recursive-shock LP.

LP base origins are identical to VAR dates at impact; available leads reduce supported origins by horizon, from 247 at impact. Dates/counts appear in the IRF exports. This standard horizon restriction does not permit outcomes beyond the frozen comparison endpoint.

4.2 Türkiye: ordering sensitivity and imprecision

Türkiye: the same estimates, three identification comparisons



100 bp tightening: darker proxy 68/90% bands. Dashed sign ranges are not confidence bands and may lie outside the view. Y-limits use non-sign estimates/90% bands. Proxy ends 2025-10; others 2026-07. ECB/Eurostat/CBRT/FRED; author calculations.

Figure 2: Türkiye recursive, FX-first and local-projection responses. Price responses accumulate the baseline monthly-inflation outcome. Intervals remain wide; exchange-rate increases indicate depreciation.

Table 4: Methods comparison on a common sample and lag specification, month twelve

p	Method	Outcome	Response	Lower	Upper	Type
3	recursive VAR	activity	0.351	-2.187	1.977	90%
3	recursive VAR	price	0.517	-0.600	1.309	90%
3	recursive VAR	fx	3.400	-0.101	6.807	90%
3	recursive-shock LP	activity	2.969	-0.929	6.866	90%
3	recursive-shock LP	price	0.635	-0.341	1.612	90%
3	recursive-shock LP	fx	-0.107	-4.879	4.665	90%
3	sign-restricted VAR	activity	0.105	-19.927	28.738	range
3	sign-restricted VAR	price	-2.813	-20.480	6.655	range
3	sign-restricted VAR	fx	4.821	-41.383	53.246	range
3	proxy VAR	activity	-0.403	-2.098	1.071	90%
3	proxy VAR	price	0.172	-2.316	1.895	90%
3	proxy VAR	fx	4.551	1.889	7.630	90%
7	recursive VAR	activity	-0.017	-2.481	1.808	90%
7	recursive VAR	price	0.068	-1.175	0.999	90%
7	recursive VAR	fx	2.465	-1.072	5.728	90%
7	recursive-shock LP	activity	2.725	-0.952	6.402	90%
7	recursive-shock LP	price	-0.070	-1.211	1.071	90%
7	recursive-shock LP	fx	-0.866	-7.018	5.285	90%
7	sign-restricted VAR	activity	-2.420	-25.599	13.274	range
7	sign-restricted VAR	price	-4.083	-21.806	5.834	range
7	sign-restricted VAR	fx	4.234	-33.383	51.583	range
7	proxy VAR	activity	-0.587	-2.201	0.901	90%
7	proxy VAR	price	-0.078	-3.132	2.392	90%
7	proxy VAR	fx	3.305	0.648	5.505	90%

Source: v3_methods_comparison.csv. April2005–October2025;247 usable VAR observations at each lag. Identical variables/transforms/deterministics and100bp normalization. Recursive-shock LP changes response estimator, not its shock identification;235 supported origins at horizon12. Sign ranges are not statistical intervals. Proxy inference is conditional, F3=19.598, F7=18.826; neither lag selected for significance.

Table 5: Türkiye responses at the headline horizon

Method	Outcome	Response	Lower 90%	Upper 90%
recursive	activity	0.011	-0.505	0.336
recursive	price	-0.288	-0.851	0.574
recursive	fx	-0.553	-1.457	0.879
LP HAC	activity	-1.054	-1.899	-0.210
LP HAC	price	-0.183	-1.011	0.645
LP HAC	fx	0.325	-0.715	1.364
FX first	activity	-0.034	-0.530	0.324
FX first	price	-0.349	-1.008	0.501
FX first	fx	-0.640	-1.692	0.828

Source: monetary-response aggregates, horizon 12. Price responses accumulate monthly inflation; exchange-rate increases mean depreciation.

The baseline recursive price response is -0.288 with interval [-0.851, 0.574]. The FX-first and LP estimates also have negative headline point estimates but include zero. This is compatible with contractionary price transmission and also with weakly identified dynamics. LP changes the propagation estimator; it inherits the shock’s identifying assumptions.

The effective funding indicator and the long announced-rate comparator represent different monetary-policy operations. Their difference should not be interpreted as alternative noisy measurements of one invariant policy regime. The current full-sample VAR remains unstable by the companion criterion (radius 1.01536). The price transformation addresses our transformation criterion but does not remove this limitation. No puzzle is removed through ex post ordering or lag tuning.

4.3 Variance accounting and historical inflation

Table 6: Recursive policy-shock forecast-error variance shares

Area	Outcome	Share	Lower 90%	Upper 90%
EA	activity	0.032	0.012	0.130
EA	price	0.002	0.001	0.053
EA	fx	0.063	0.015	0.185
TR	activity	0.022	0.012	0.076
TR	price	0.031	0.019	0.111
TR	fx	0.019	0.008	0.111

Recursive Cholesky; EA usable 2005-04–2026-07, p=7; TR usable 2011-10–2026-07, p=9. Horizon12; fractions, not percentages; EA price level and TR inflation-native FEVD are different outcomes.

For recursive forecast-error variance attribution at horizon H , we compute

$$FEVD_{i,j}(H) = \frac{\sum_{h=0}^{H-1} (\Phi_h L)_{ij}^2}{\sum_k \sum_{h=0}^{H-1} (\Phi_h L)_{ik}^2}. \quad (10)$$

The attribution uses orthogonal unit-variance innovations, not the normalized policy shock used to display IRFs. Türkiye’s FEVD describes inflation-native forecast errors; accumulation for a price-level response does not change that FEVD outcome. We compute historical contributions using

Table 7: Recursive Cholesky: annual averages of monthly log inflation and policy contributions

Area	Year	Months	Monthly infl.	Policy part	Lower90%	Upper90%
EA	2020	12	-0.023	0.004	-0.080	0.101
EA	2022	12	0.738	0.001	-0.095	0.088
EA	2025	12	0.163	-0.017	-0.120	0.114
EA	2026 partial	7	0.367	-0.023	-0.134	0.126
TR	2020	12	1.136	-0.096	-1.120	1.309
TR	2022	12	4.136	-0.308	-2.098	1.797
TR	2025	12	2.243	-0.355	-2.782	3.103
TR	2026 partial	7	2.588	0.090	-3.226	3.964

EA usable 2005-04–2026-07,p=7; TR usable 2011-10–2026-07,p=9. Units: percentage points of monthly100delta(log P), arithmetic annual mean, NOT annual inflation.2026: January–July,7 months, no annualisation. Partial first/final years are not full-year averages. Bands apply to policy contributions; deterministic/initial history and all other shocks complete the identity.

$y_t = y_t^{\text{init,det}} + \sum_j c_{j,t}$ with $c_{j,t} = Le_j \epsilon_{j,t} + \sum_\ell A_\ell c_{j,t-\ell}$. For level prices, we difference contributions to express monthly inflation; for Türkiye’s native inflation they are already monthly inflation. Conditional deterministic and initial-history components complete the accounting identity. We

report arithmetic annual averages of monthly $100\Delta \log P$, in percentage points: an annual mean of monthly log inflation, not annual inflation or cumulative inflation. We difference already 100log-scaled EA price contributions once, with no further multiplication; TR native monthly inflation is already on the same scale. Every year records contributing months and actual coverage. The final year is January–July only and is not annualised. Initial/deterministic plus all shock components reconstruct the represented level and monthly series; checks are exported separately.

Historical accounting is descriptive conditional on the structural ordering. It is not a causal attribution of each inflation episode, and the bootstrap applies parameter draws to observed history rather than replaying an alternative policy sequence. Annual averaging also hides within-year reversals. Forecast-error shares, historical contributions and impulse responses answer distinct questions and should not be used interchangeably.

5 Robustness

Table 8: Specifications and alternative estimates

Declared alternative	Sample	Lag	Price	90% interval
TR announced	2006-01 / 2026-07	6	-1.326	[-2.033, -0.311]
EA overnight long	1999-12 / 2019-12	7	-0.429	[-1.393, 0.664]
EA bundesbank long	1999-12 / 2026-07	3	0.519	[-0.351, 1.162]
EA overnight common	2004-09 / 2026-07	4	0.000	[-1.852, 1.584]
EA core	2004-09 / 2026-07	3	0.573	[-0.061, 0.973]
TR core b	2011-01 / 2026-07	2	-1.206	[-1.827, -0.432]
TR core c	2011-01 / 2026-07	2	-1.267	[-1.819, -0.549]
EA growth	2004-09 / 2026-07	4	0.037	[-1.019, 1.368]
TR growth	2011-01 / 2026-07	6	-1.178	[-3.281, 0.564]
TR levels comparison	2011-01 / 2026-07	2	-0.984	[-1.642, -0.131]
EA BIC	2004-09 / 2026-07	1	-0.021	[-0.707, 0.221]
TR BIC	2011-01 / 2026-07	2	-1.124	[-1.821, -0.291]
TR FX first	2011-01 / 2026-07	9	-0.349	[-1.008, 0.501]
TR easing dummy	2011-01 / 2026-07	2	-0.392	[-0.990, 0.199]
EA LP	See method sample	–	-0.081	[-1.157, 0.996]
TR LP	See method sample	–	-0.183	[-1.011, 0.645]
EA sign set	See method sample	–	-3.508	range [-20.017, 5.181]
TR sign set	See method sample	–	-1.531	range [-13.522, 4.641]
EA event proxy	See method sample	–	0.113	[-2.320, 2.084]
EA ordinary block	See method sample	–	0.262	[-0.978, 1.070]
TR ordinary block	See method sample	–	-0.288	[-0.988, 0.492]
FX pass-through	See method sample	–	1.142	[-1.339, 3.622]
Recent TR LP, native	See method sample	–	–	[-, -]
Early TR regime	2011-01 / 2018-05	1	-0.716	[-1.167, -0.393]
Middle TR regime	2018-06 / 2023-05	2	-1.665	[-4.566, 1.751]
Johansen/VECM	Integration conditions	–	–	not estimated
TR bond-yield indicators	Official source unavailable	–	–	not included
Old-base core splice	Bridge tolerance	–	–	rejected

Sources: monetary-response aggregates and earlier regime tables. Responses at horizon 12; sign endpoints are ranges. Recent LP reports native inflation; pass-through scales the FX innovation, not the policy shock. Rejected/unestimated alternatives have no fabricated numerical result.

We report every specified alternative, including earlier regime estimates, recent native-inflation local projections, incompatible index splicing and unmet integration conditions for a VECM. Sign extrema are ranges, not intervals. Pass-through uses a recursive FX-innovation LP scaled by $100\log(1.1)$, an exact ten-percent depreciation; it is not a cointegrating elasticity. Unavailable

official Türkiye bond-yield indicators remain out of scope.

6 Limitations and future work

Within each new common-sample lag panel the variables, sample, transformations, deterministic terms and shock scale are held constant. The response estimator also changes for LP, so this is a methods comparison, not an identification-only experiment. Sign-admissible ranges condition on the reduced form, while sampling intervals have different inference assumptions. Inherited robustness and TR results retain their original windows and remain separately labelled.

Instrument relevance is not exclusion. Event-window yield movements can include central-bank information and concurrent financial news. Conventional first-stage inference may be fragile under serial dependence, and the conditional proxy bands exclude weak draws rather than supplying identification-robust coverage. Fixed slopes omit one component of sampling variation.

Unit roots, breaks and unstable finite-sample companion estimates complicate conventional uncertainty. The euro-area radius is 1.00388 and the Türkiye radius 1.01536. Small-sample correction does not forcibly stabilize either baseline. Finite-horizon responses can still be computed, but long-horizon forecasts, variance attribution and narrow coverage interpretations require caution. Marginal residual tests show additional misspecification concerns even after the first-order screen.

Sign identification imposes early disinflation; an apparently favorable price response is therefore partly assumed. A finite random search does not exhaust a continuous identified set. Latest-vintage data do not reconstruct historical information sets, and index revisions or geographical definitions can alter comparability. Regime exercises are descriptive, with the newest Türkiye segment particularly short. No raw data or private shock series are redistributed.

7 Conclusion

Both predeclared common-sample lag panels support a conditional proxy euro-appreciation response at twelve months, whereas price and activity signs remain imprecise. The conclusion relies on instrument exogeneity and conditional inference, not merely $F > 10$. Comparing recursive VAR, sign-restricted VAR, recursive-shock LP and proxy VAR makes explicit which shock restriction and response estimator support each statement. Türkiye's inherited negative price point estimates remain imprecise and unstable dynamics qualify both applications.

Recursive FEVD and historical accounting are conditional descriptive allocations, not proxy-identified inflation explanations or causal policy counterfactuals. Future work includes information-shock separation and alternative windows, dependence-robust strength and weak-IV/full-chain inference, persistence-robust LP, break/shrinkage/model-size diagnostics, TR intervention comparability, sign-grid sensitivity and economically interpretable counterfactual paths. None is implemented or selected on significance here.

References

- Carlo Altavilla, Luca Brugnolini, Refet S. Gürkaynak, Roberto Motto, Giuseppe Ragusa (2019). Measuring euro area monetary policy. *Journal of Monetary Economics* 108, 162-179. [10.1016/j.jmoneco.2019.08.016](https://doi.org/10.1016/j.jmoneco.2019.08.016).
- Òscar Jordà (2005). Estimation and Inference of Impulse Responses by Local Projections. *American Economic Review* 95(1), 161-182. [10.1257/0002828053828518](https://doi.org/10.1257/0002828053828518).

- Lutz Kilian (1998). Small-sample Confidence Intervals for Impulse Response Functions. *Review of Economics and Statistics* 80(2), 218-230. [10.1162/003465398557465](#).
- Christopher A. Sims, James H. Stock, Mark W. Watson (1990). Inference in Linear Time Series Models with some Unit Roots. *Econometrica* 58(1), 113–144. [10.2307/2938337](#).
- Harald Uhlig (2005). What are the effects of monetary policy on output? Results from an agnostic identification procedure. *Journal of Monetary Economics* 52(2), 381-419. [10.1016/j.jmoneco.2004.05.007](#).

Table 9: Saved computation and accounting diagnostics

Area	Radius	Block ok	Unstable	Signs	Attempts	HD error
EA	1.00388	499	444	1000	4074	3.14e-11
TR	1.01536	499	484	1000	3262	1.14e-11

Source: author’s calculations from the documented inputs. Block ok counts successful ordinary block resamples, not Kilian corrections. HD error is the maximum absolute reconstruction error.

A Lag selection and baseline conventions

Table 10: Common-window information criteria before the residual screen

Area	Lag	Common n	AIC	BIC	HQ
EA	1	251	-2.807	-1.290	-2.197
EA	2	251	-3.080	-1.057	-2.266
EA	3	251	-3.001	-0.472	-1.983
EA	4	251	-2.957	0.077	-1.736
EA	5	251	-2.858	0.681	-1.434
EA	6	251	-2.822	1.224	-1.194
EA	7	251	-2.803	1.748	-0.972
EA	8	251	-2.620	2.436	-0.586
EA	9	251	-2.551	3.012	-0.312
EA	10	251	-2.513	3.554	-0.072
EA	11	251	-2.485	4.088	0.160
EA	12	251	-2.416	4.663	0.433
TR	1	175	3.774	5.962	4.662
TR	2	175	2.029	4.886	3.188
TR	3	175	1.941	5.467	3.371
TR	4	175	1.967	6.163	3.669
TR	5	175	1.987	6.852	3.960
TR	6	175	1.971	7.504	4.215
TR	7	175	2.110	8.313	4.626
TR	8	175	2.209	9.081	4.996
TR	9	175	2.253	9.794	5.312
TR	10	175	2.398	10.608	5.728
TR	11	175	2.358	11.238	5.960
TR	12	175	2.317	11.866	6.190

Source: lag-selection aggregates. Our final lag selection adds the equation-BG residual screen; BIC-only diagnostics are reported separately.

Information criteria compare a common window; the residual screen is an additional declared choice, not an information criterion. We retain the complete residual-screen search paths, including candidates that fail the screen and instances in which we retain the BIC choice. Model-specific maximum feasible lag respects rank and residual degrees of freedom. We distinguish the BIC-only specifications from the final models selected with the additional residual screen.

B Residual diagnostics

We use vector residual resampling with fixed lag, seasonal terms, transformation and seed 20261007. Kilian-style and ordinary-block simulations have distinct dependence and correction rules. Historical identities reconstruct to the reported numerical precision, but arithmetic reconstruction does not validate economic identification. Proxy acceptance counts are reported in the main text. We report the resampling design and its recorded completion statistics.

Table 11: Marginal residual diagnostics at the selected lag

Area	Equation	BG(1) p	BG(6) p	BG(12) p	JB p
EA	brent	0.278	0.422	0.135	0.000
EA	vix	0.928	0.092	0.019	0.000
EA	activity	0.603	0.436	0.298	0.000
EA	price	0.470	0.034	0.003	0.000
EA	policy	0.505	0.090	0.033	0.000
EA	fx	0.735	0.395	0.204	0.169
TR	brent	0.080	0.601	0.186	0.000
TR	vix	0.537	0.039	0.077	0.000
TR	fedfunds	0.053	0.239	0.115	0.000
TR	activity	0.069	0.000	–	0.213
TR	price	0.450	0.000	–	0.000
TR	policy	0.208	0.002	–	0.000
TR	fx	0.653	0.017	–	0.000

Source: author’s calculations from the documented inputs. Unavailable tests are dashes. These are equation tests, not an omnibus system LR test; normality failures motivate transparent bootstrap qualifications.

C Integration and rebasing diagnostics

Table 12: Saved log-CPI and first-difference integration diagnostics

Sample	Transform	Test	Statistic	p value	Lag
2006-01:2026-07	level	ADF c	3.215	1.000	3
2006-01:2026-07	level	KPSS c	2.050	0.010	10
2006-01:2026-07	level	PP c	4.554	1.000	16
2006-01:2026-07	level	ZA ct	-4.033	0.481	3
2006-01:2026-07	first_difference	ADF c	-2.785	0.060	5
2006-01:2026-07	first_difference	KPSS c	1.496	0.010	8
2006-01:2026-07	first_difference	PP c	-8.996	0.000	16
2006-01:2026-07	first_difference	ZA ct	-9.373	0.001	2
2011-01:2026-07	log_level	ADF c	2.362	0.999	3
2011-01:2026-07	log_level	KPSS c	1.776	0.010	9
2011-01:2026-07	log_level	PP c	3.519	1.000	15
2011-01:2026-07	log_level	ZA ct	-3.548	0.788	1
2011-01:2026-07	first_difference	ADF c	-4.023	0.001	2
2011-01:2026-07	first_difference	KPSS c	1.216	0.010	8
2011-01:2026-07	first_difference	PP c	-7.700	0.000	15
2011-01:2026-07	first_difference	ZA ct	-10.245	0.001	0

Source: author’s calculations from the documented inputs. ADF/PP null: unit root; KPSS null: stationarity. Reported KPSS endpoint p values are bounds from lookup tables. ZA allows one endogenous break.

We assess the integration order of log CPI, nominal FX and REER over its documented longer Türkiye window. We require agreement across ADF, PP and KPSS before estimating a VECM; it is not proof of integration order. The baseline price switch also responds to levels stability, so failure of the I(2) rule alone would not prevent that switch. The narrower baseline diagnostic rows remain separately labelled. Old/current core-index growth disagreement preserves our use of unspliced current-base indices. Current indices cover the target period, so the alternative price responses do not require a synthetic bridge. A failed tolerance check is a finding, not an instruction to search for a rescaling that passes.